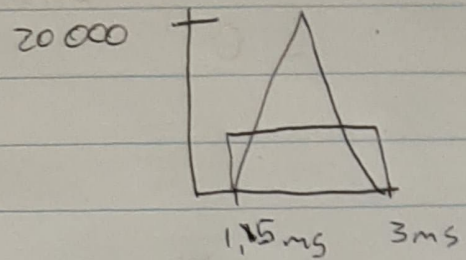


FSK Online Test 2

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$$\begin{aligned} \textcircled{2} a) \quad & \frac{1}{2} b h \\ &= \frac{1}{2} (1,5 \times 10^{-3}) (20000) \\ &= 15 \end{aligned}$$

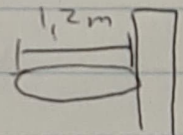


$$\begin{aligned} b &= 3 - 1,5 \\ &= 1,5 \times 10^{-3} \end{aligned}$$

$$\begin{aligned} b) \quad & L \times B = 15 \\ (1,5 \times 10^{-3}) \times B &= 15 \\ \therefore F &= 10000 \text{ N} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad I &= F \Delta t \\ +28824 &= \end{aligned}$$

$$\begin{aligned} I &= m \Delta v \\ &= (201)(0 - 24) \\ &= -28824 \text{ Ns} \end{aligned}$$



$$\begin{aligned} m &= 1,201 \text{ kg} \\ v_i &= 24 \text{ m/s} \\ \Delta t &= ? \end{aligned}$$

$$\Delta x = \left(\frac{v_f + v_i}{2} \right) \Delta t$$

$$1,2 = \left(\frac{12}{2} \right) \Delta t$$

$$\therefore t = 0,1 \text{ s}$$

$$288240$$

$$\begin{aligned} \textcircled{4} a) \quad \vec{I} &= \frac{1}{2} b h + \frac{1}{2} b h + (L \times B) \\ &= 4 + 4 + 4 \\ &= 12 \text{ N.s} \end{aligned}$$

$$\begin{aligned} b) \quad I &= m \Delta v \\ 12 &= 2,7 (v_f) \\ 4,44 \text{ m/s} &= v_f \end{aligned}$$

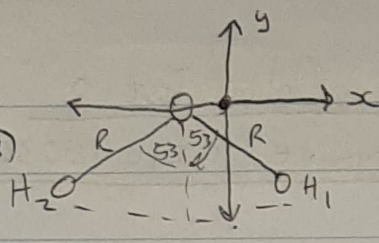
$$v_i = 0$$

$$\begin{aligned} d) \quad F_{\text{ave}} &: L \times B = 12 \\ 5 \times B &= 12 \\ B &= 2,4 \\ \therefore F &= 2,4 \text{ N} \end{aligned}$$

$$\begin{aligned} c) \quad 12 &= 2,7 (v_f + 2,4) \\ v_f &= 2,04 \text{ m/s} \end{aligned}$$

$$⑤ C_{mx} = \frac{\sum m_i x_i}{\sum m_i}$$

$$= \frac{(0,083)(1,008) - (0,083)(1,008)}{2,016} = 0$$



$$\sin \theta = \frac{x}{R}$$

$$\sin 53 = \frac{x}{0,104}$$

$$x = \frac{0,104 \cdot \sin 53}{1} = 0,083 \text{ nm}$$

$$H_1 (0,083; 0,063)$$

$$H_2 (-0,083; 0,063) \cos 53 = \frac{y}{2}$$

$$y = 0,063 \text{ nm}$$

$$C_{my} = \frac{\sum m_i y_i}{\sum m_i}$$

$$= \frac{0,127008}{18,0159}$$

$$=$$

$$0,007,049$$

⑥ a)

$$E_o = E_f$$

$$mgh = \frac{1}{2} mv^2$$

$$441 = 7,5v^2$$

$$v_i = 7,668 \text{ m/s}$$

$$⑦ a) \alpha = \frac{\Delta \omega}{\Delta t}$$

$$-2,04 = \frac{0 - 12,5664}{\Delta t}$$

$$\Delta t = \frac{-12,5664}{-2,04}$$

$$= 6,16 \text{ s}$$

$$\omega = 12,5664 \text{ rad/s}$$

$$b) \theta = \theta_o + \omega_o t + \frac{1}{2} \alpha (t)^2$$

$$= (12,5664)(6,16) + \frac{1}{2}(-2,04)(6,16)^2$$

$$= 38,70 \text{ rad}$$

$$(8) a) 9,177 = \sqrt{\frac{5}{m} (0,1^2 - 0,05^2)}$$

$$4,1772 = \frac{5}{m}$$

$$m = \frac{5}{4,1772}$$

$$= 1,20$$

$$V = \cancel{17,7} \text{ m/s } 0,177 \text{ m/s}$$

$$A = 0,1 \text{ m}$$

$$K = 5$$

$$a(x) = \frac{-k}{m} x$$

$$= \frac{-5}{1,2} (0,05)$$

$$= -0,21$$

$$(10) \cancel{v(x) = \sqrt{8}}$$

$$8 \cos\left(t + \frac{\pi}{8}\right)$$

$$8 \sin\left(t + \frac{\pi}{8}\right) \times 1$$

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